COSC 4370 - Homework 2

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1 Problem

This homework requires creating four scenes using OpenGL. The first three scenes reproduce the provided examples: eight teapots arranged around a center, a stepped pyramid made from cubes, and an inverted triangle of teapots. The fourth scene is an original design. The program uses the provided *main.cpp* file and the four functions *problem1*, *problem2*, *problem3*, and *problem4*.

The original scene must include a triangle specified directly with coordinates and a meaningful use of nested *glPushMatrix* calls. Each scene can be selected while the program is running by pressing its corresponding number key.

2 Method

The program uses OpenGL transformations to place and resize objects rather than defining every object from individual vertices. *glTranslatef* changes an object's position, *glRotatef* changes its orientation, and *glScalef* changes its dimensions. *glPushMatrix* and *glPopMatrix* save and restore transformations, allowing objects to be drawn without affecting the next object.

For the first and third scenes, *glutSolidTeapot* is used to create the teapots. The second scene uses *glutSolidCube* for each level of the pyramid. The final scene uses a triangle fan with manually specified vertices to form a heart, along with scaled cubes for its stand and base.

3 Implementation

3.1 Problem 1 - Circular Teapot Arrangement

A for loop draws eight teapots. Each repetition rotates the coordinate system by a multiple of 45 degrees and then translates a teapot away from the center. The four teapots on the main horizontal and vertical axes use a distance of 1.0, while the four diagonal teapots use 0.75. This brings the diagonal teapots inward to better match the supplied example. An additional 90-degree rotation turns the teapots inward, and each teapot is drawn with a size of 0.10.

3.2 Problem 2 - Stepped Pyramid

A for loop creates eleven layers with *glutSolidCube*. Each successive layer is moved upward by 0.15 units. The variable *size* begins at 2.5 and decreases by 0.23 for every layer. This value is used for scaling in both the X and Z directions, while the Y dimension stays at 0.15. The resulting cubes are stacked into a square stepped pyramid that becomes narrower toward the top.

3.3 Problem 3 - Inverted Teapot Triangle

Nested for loops generate six rows of teapots. The top row contains six teapots, and each following row contains one fewer, ending with one teapot at the bottom. This gives a total of 21 teapots. The horizontal position is calculated using the number of teapots in each row so that every row stays centered. The final horizontal spacing is 0.37, the vertical spacing is 0.30, and the teapot size is 0.12.

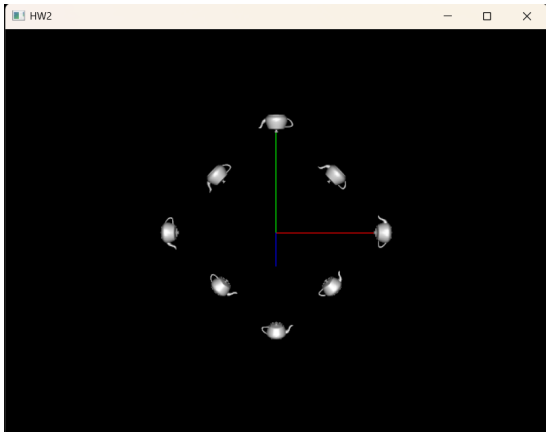
3.4 Problem 4 - Heart Trophy

The original scene is a white heart displayed above a rectangular pedestal. The heart outline is specified directly with *glVertex3f* coordinates inside *GL_TRIANGLE_FAN*. The triangle fan connects these outline points to a central vertex to fill the heart shape. A scaled cube forms the pedestal and a narrower scaled cube forms the support between the base and the heart.

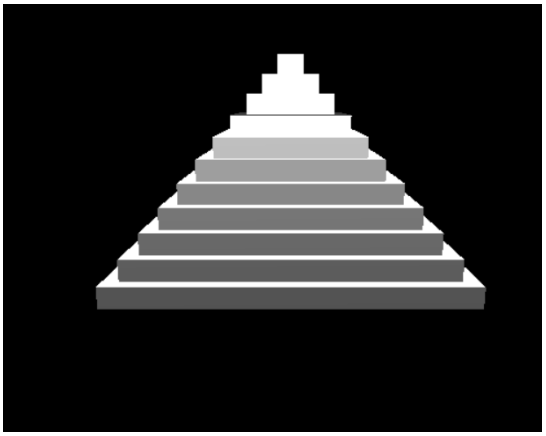
Nested *glPushMatrix* and *glPopMatrix* calls are used to apply an overall translation to the scene, followed by separate translations and scaling for the heart and the pedestal components. This allows the parts to move together while keeping their individual sizes and positions independent. The heart retains the white appearance used in the final scene.

4 Results

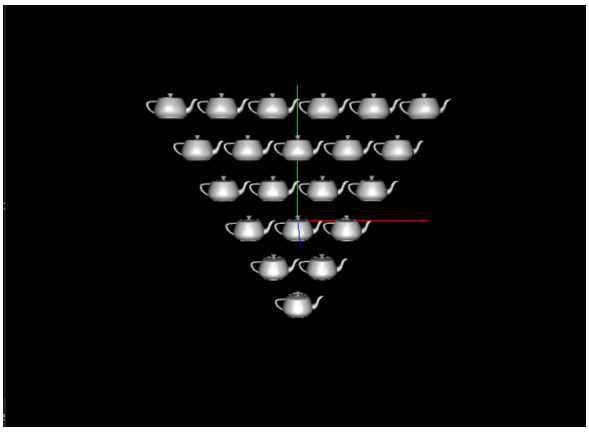
All four functions produced visible scenes in the OpenGL window. Problem 1 produced eight inward-facing teapots with the diagonal positions moved closer to the center. Problem 2 produced an eleven-layer square pyramid. Problem 3 produced six centered rows containing 21 teapots. Problem 4 produced a white heart on a stand with a rectangular base. The final results reproduce the first three references approximately and satisfy the required techniques for the original scene.



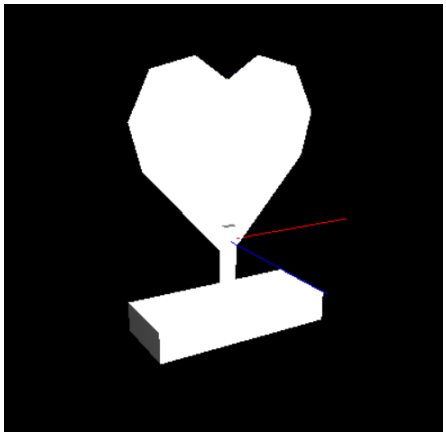
Problem 1 - Eight teapots



Problem 2 - Pyramid



Problem 3 - Inverted triangle



Problem 4 - Heart trophy